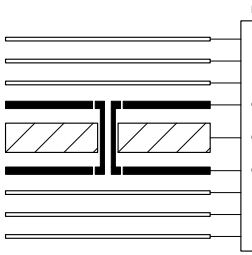
JONES_MIXER Fabrication Document

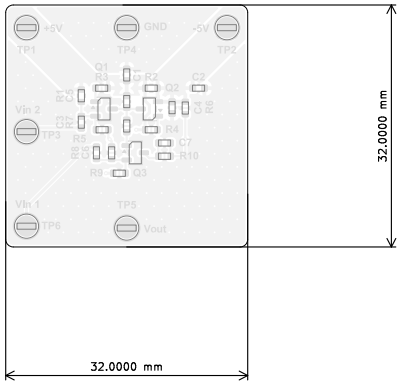
Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	L1	0.035mm (1.00oz)		Signal	GBR
Core		1.471mm	FR4_7628	Dielectric	
Copper	L2	0.035mm (1.00oz)		Signal	GBR
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.581mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



Impedance Table

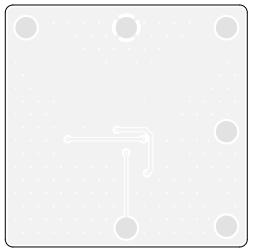
Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 7

JONES_MIXER Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

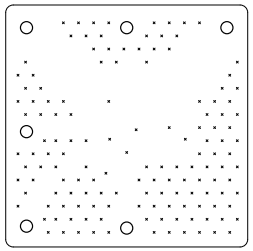
blue devil MOTORSPORTS	Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea
		Board Name: JONES_MIXER		Project Name: Jones Mixer	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 7

JONES_MIXER Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	142	0.25mm (9.84mils)	PTH	Round	L1 - L2	Via
○	6	1.60mm (62.99mils)	PTH	Round	L1 - L2	Pad
	Total 148					



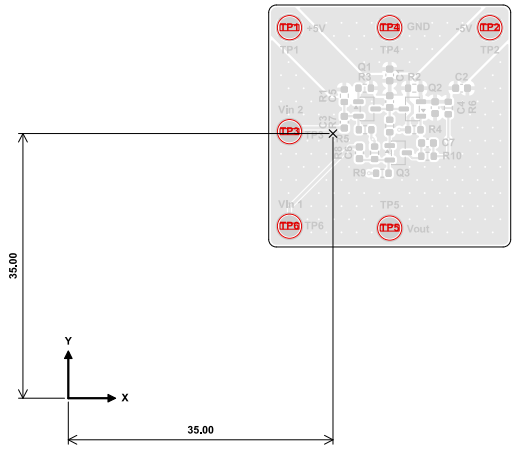
Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea	
	Board Name: JONES_MIXER		Project Name: Jones Mixer		
Sheet Title: Drill Drawing (L1 - L2)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1	
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 7	

JONES_MIXER Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	+5V	2.75	29.00
TP2	-5V	29.25	29.00
TP3	Vi2	2.75	15.25
TP4	GND	16.00	29.00
TP5	Vo	16.00	2.50
TP6	Vi1	2.75	2.75
TP7		-140.00	105.00
TP8		-140.00	105.00
TP9		-140.00	105.00
TP10		-140.00	105.00
TP11		-140.00	105.00
TP12		-140.00	105.00

Ref.	Net	X [mm]	Y [mm]



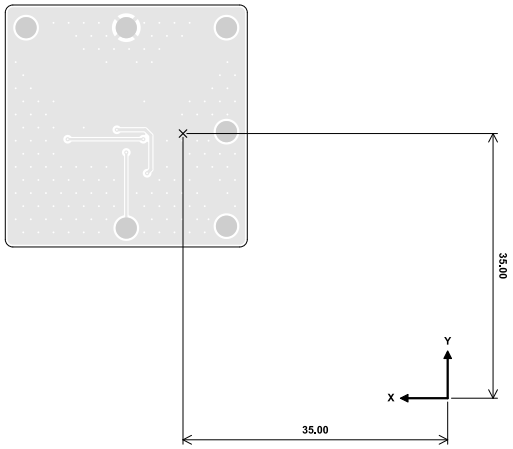
All dimensions are in millimeters unless otherwise specified.

blue devil MOTORSPORTS	Comments:	Company: LTU	Variant: RELEASED	Git Hash: e165eea
		Board Name: JONES_MIXER	Project Name: Jones Mixer	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13 / Revision: 1.0.1
	Sheet Path:		Reviewer:	Size: A4 / Sheet: 4 of 7

JONES_MIXER Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

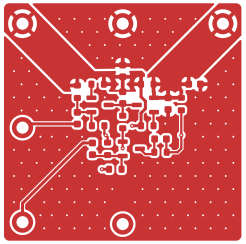


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 7

JONES_MIXER Fabrication Document

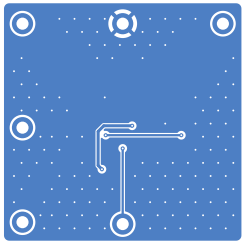
L1 (Scale 1:1)



Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: L1 (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 7

JONES_MIXER Fabrication Document

L2 (Scale 1:1)



Comments:	Company: LTU		Variant: RELEASED	Git Hash: e165eea
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: L2 (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: A.B., C.B., A.V.	Date: 2024-04-13	Revision: 1.0.1
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 7